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Assignment2

University College London (2015)

“Data are facts, observations or experiences on which an argument or theory is constructed or tested. Data may be numerical, descriptive, aural or visual. Data may be raw, abstracted or analyzed, experimental or observational. Data include but are not limited to: laboratory notebooks, field notebooks, primary research data (including research data in hardcopy or in computer readable form), questionnaires, audiotapes, videotapes, models, photographs, films, test responses. Research collections may include slides, artefacts, specimens, samples.”

Strength: It takes into consideration a diverse range of data types.

Weakness: It could be more specific about data management.

EPSRC

Research data are the raw materials collected, processed and studied in the undertaking of research. They are the evidential basis that substantiates published research findings. They may be primary data generated or collected by the researcher, or secondary data collected from existing sources and processed as part of the research activity. In addition to the 'raw' data, research data include information about the means necessary to generate data or replicate results, such as computer code, experimental methods and instruments used, and essential interpretive and contextual information, e.g. specifications of variables. The raw data of research may exist in digital and non-digital formats.

Strength: It categorize digital and non-digital formats of the data.

Weakness: lack of specificity

Pryor (2012)

The output from any systematic investigation involving a process of observation, experiment or the testing of a hypothesis, which when assembled in context and interpreted expertly will produce new knowledge.

Strength: the generation of new knowledge

Weakness: It doesn't clearly define research data.

Assignment3

Chosen data portal: Obis

- **Data Management:** OBIS effectively manages and organizes scientific data related to ocean biogeography. It supports various data formats, including species occurrence data, environmental data, and sampling protocols. The portal ensures data quality by promoting data standardization and following best practices for data management.
- **Metadata:** OBIS places a strong emphasis on metadata to enhance data understanding and interoperability. The portal requires comprehensive metadata for all datasets, including essential information such as taxonomic classification, geographic location, sampling methods, and data quality indicators. By adhering to well-defined metadata standards, OBIS ensures that datasets are properly described and can be easily integrated with other scientific datasets.
- **Data Access:** OBIS provides users with seamless access to its extensive collection of ocean biogeographic data. Through the user-friendly interface, users can search for datasets using various criteria, including taxonomic names, geographic regions, and time periods. The portal offers multiple download options, allowing users to obtain the data in common formats such as CSV or Excel. Additionally, OBIS provides an API (Application Programming Interface) that enables programmatic access to the data, facilitating integration with other tools and platforms.
- **Data Discovery:** OBIS excels in supporting data discovery through its robust search functionality. Users can perform keyword searches, apply taxonomic filters, and narrow down results using geographic filters. The portal also offers advanced search features, such as spatial and temporal queries, enabling users to find specific datasets of interest. The search interface is user-friendly, making it easy to navigate and refine search results.
- **Data Analysis:** While OBIS focuses primarily on data aggregation and dissemination, it provides links to external tools and resources for data analysis, visualization, and transformation. These external tools can be leveraged by users to perform in-depth analysis and visualization of the data obtained from OBIS. The portal serves as a valuable starting point for researchers to access relevant data for their analyses.
- **Data Sharing:** OBIS actively promotes data sharing and collaboration among the scientific community. The portal allows users to contribute their datasets, expanding the breadth and depth of available data. OBIS facilitates data sharing through its Data Partners Network, where researchers can collaborate on specific projects and contribute their expertise. Additionally, OBIS

supports data citation, ensuring proper attribution for shared datasets and promoting the recognition of data contributors.